WORKED SOLUTION

STEP III 1993, Question 10

Pure • Matrix Transformations

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Read the question before you go on. It is in the STEP Support Programme's question database:

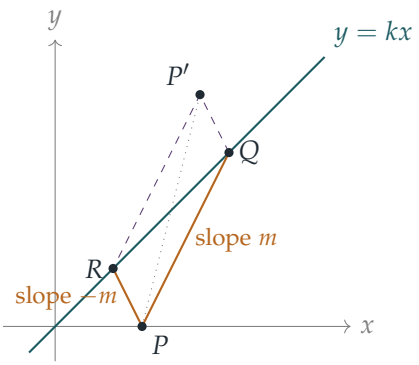
step.maths.org/questions/93-s3-q10

The key idea

Everything here is the intersection of two straight lines, so the coordinates come from solving one linear equation at a time. The parallelogram gives P' with no further work: in a parallelogram $PQP'R$ the diagonals bisect each other, so $P + P' = Q + R$ as position vectors. Once P' is written as a matrix times P, the area question is just a determinant.

Solution

Throughout we need $m^2 \neq k^2$: if $m = \pm k$, one of the two lines through P is parallel to $y = kx$ and never meets it (or lies along it), so the construction fails.



(The picture uses $m = 2$, $k = 1$ and $P = (1, 0)$.)

The point Q

The line through $P(x_1, y_1)$ parallel to $y = mx$ is $y - y_1 = m(x - x_1)$. At Q we also have $y = kx$, so

$$kx - y_1 = mx - mx_1 \quad \Longrightarrow \quad (m-k)x = mx_1 - y_1 \quad \Longrightarrow \quad x = \frac{mx_1 - y_1}{m-k},$$

dividing by $m-k \neq 0$. Then $y = kx$ gives

$$Q = \left(\frac{mx_1 - y_1}{m-k},\ \frac{k(mx_1 - y_1)}{m-k} \right),$$

as required.

The point R

The line through P parallel to $y = -mx$ is the same calculation with m replaced by $-m$:

$$x = \frac{-mx_1 - y_1}{-m-k} = \frac{mx_1 + y_1}{m+k}, \qquad R = \left(\frac{mx_1 + y_1}{m+k},\ \frac{k(mx_1 + y_1)}{m+k} \right).$$

The point P'

> **Ask yourself** How do we find the fourth vertex of a parallelogram without any more intersecting lines?

The vertices go round in the order P, Q, P', R, so PP' and QR are the diagonals. The diagonals of a parallelogram bisect each other, so they have the same midpoint:

$$\tfrac{1}{2}(P + P') = \tfrac{1}{2}(Q + R) \quad \Longrightarrow \quad P' = Q + R - P.$$

(Equivalently, $\overrightarrow{QP'} = \overrightarrow{PR}$.) Both Q and R lie on $y = kx$, so it is easiest to add their x-coordinates first. Over the common denominator $m^2 - k^2$,

$$\begin{aligned} x_Q + x_R &= \frac{(mx_1 - y_1)(m+k) + (mx_1 + y_1)(m-k)}{m^2 - k^2} \\ &= \frac{(m^2x_1 + mkx_1 - my_1 - ky_1) + (m^2x_1 - mkx_1 + my_1 - ky_1)}{m^2 - k^2} \\ &= \frac{2(m^2x_1 - ky_1)}{m^2 - k^2}, \end{aligned}$$

and $y_Q + y_R = k(x_Q + x_R)$. Therefore, writing $P' = (x', y')$,

$$\begin{aligned} x' &= \frac{2(m^2x_1 - ky_1)}{m^2 - k^2} - x_1 = \frac{(m^2 + k^2)x_1 - 2ky_1}{m^2 - k^2}, \\ y' &= \frac{2k(m^2x_1 - ky_1)}{m^2 - k^2} - y_1 = \frac{2km^2x_1 - (m^2 + k^2)y_1}{m^2 - k^2}. \end{aligned}$$

T as a matrix

Both coordinates are linear in x_1, y_1 with no constant term, so

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \mathbf{M} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \qquad \mathbf{M} = \frac{1}{m^2 - k^2} \begin{pmatrix} m^2 + k^2 & -2k \\ 2km^2 & -(m^2 + k^2) \end{pmatrix}.$$

Areas

A linear map with matrix $\mathbf{M}$ multiplies every area by $|\det \mathbf{M}|$. Taking the factor $\frac{1}{m^2-k^2}$ out of each row,

$$\det \mathbf{M} = \frac{-(m^2+k^2)^2 + 4k^2m^2}{(m^2-k^2)^2} = \frac{-(m^4 - 2m^2k^2 + k^4)}{(m^2-k^2)^2} = \frac{-(m^2-k^2)^2}{(m^2-k^2)^2} = -1.$$

So $|\det \mathbf{M}| = 1$ and every region is mapped to a region of the same area, as required. The minus sign tells us that T reverses orientation, as a reflection does: a figure traced anticlockwise has an image traced clockwise.

? **Ask yourself** Is there a picture of what T is doing?

Yes. The midpoint of PP' equals the midpoint of QR, which lies on $y = kx$, and points of $y = kx$ are fixed (there $Q = R = P$). Solving $\mathbf{Mv} = -\mathbf{v}$ gives the row $2m^2x - 2ky = 0$, so the vectors reversed by T are the multiples of (k, m^2). Thus T is a *skew reflection*: it reflects in the line $y = kx$, but along the direction (k, m^2) rather than perpendicular to the line. Like any reflection it satisfies $\mathbf{M}^2 = \mathbf{I}$, which also fits the construction: starting from P' gives the same Q and R, and so leads back to P.

Checks

- With $m = 2$, $k = 1$, $P = (1,0)$: $Q = (2,2)$, $R = (\frac{2}{3}, \frac{2}{3})$, so $P' = (\frac{5}{3}, \frac{8}{3})$. The matrix is $\frac{1}{3}\begin{pmatrix} 5 & -2 \\ 8 & -5 \end{pmatrix}$, which sends $(1,0)$ to $(\frac{5}{3}, \frac{8}{3})$, and its determinant is $\frac{1}{9}(-25+16) = -1$.
- A point (t, kt) on the line should be fixed: $x' = \frac{(m^2+k^2)t - 2k^2t}{m^2-k^2} = t$. ✓
- The trace of $\mathbf{M}$ is 0 and its determinant is -1, so its eigenvalues are ± 1, as expected for a reflection.
- We checked all of these formulas, together with the preservation of triangle areas, exactly in rational arithmetic on about 3000 random choices of m, k and P.

Common slips

- Taking $P' = Q + R$, or $P' = P + Q + R$, instead of $Q + R - P$. The order of the letters $PQP'R$ tells you which pairs of vertices are opposite.
- Concluding from $\det \mathbf{M} = -1$ that areas become negative. The area scale factor is $|\det \mathbf{M}|$; the sign only records orientation.

Beyond STEP

A linear map that fixes a line pointwise and reverses one other direction is an *involution*, $T^2 = I$. Ordinary orthogonal reflections are the special case in which the reversed

direction is perpendicular to the line; here that happens exactly when (k, m^2) is perpendicular to $(1, k)$, that is, only when $k = 0$ (the lines $y = \pm mx$ are symmetric about the x-axis, not about $y = kx$). At university this sits inside the study of eigenvalues and diagonalisation, where T is seen as $\mathrm{diag}(1, -1)$ in the basis $(1, k)$, (k, m^2), and of the determinant as a signed volume scale factor. Strang's *Introduction to Linear Algebra* covers both ideas clearly.

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